

JKBOSE 2026

CLASS 11TH

CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

LECTURE-04 CHEMISTRY

VISHAL SIR

Bhaichara
ON
TOP



JKBose Wallah

Topics To Covered

1 MODERN PERIODIC TABLE (*Part-2*).

REVISION

6-hours + 2 hours

Backlog.

MODERN PERIODIC TABLE.

A) MODERN PERIODIC LAW.
(Z) ✓

B) FEATURES OF MODERN. P. T.
(Bohr's table).

→ GROUPS :→ 18 (1-18).

→ PERIODS :→ 7 (1-7).

1st PERIOD → 2

2nd & 3rd. → 8, 8

4th & 5th PERIOD → 18, 18

6th & 7th PERIOD. → 32, 32.

MENDELEEV'S. P. T.

A) MERITS.

SYSTEMATIC
STUDY OF
ELEMENTS

CORRECTION
OF ATOMIC
MASS.

PREDICTION
OF NEW
ELEMENTS.

B) DEFECTS.

→ POSITION OF HYDROGEN.

→ ANOMALOUS PAIR.

→ POSITION OF ISOTOPES.

→ CAUSE OF PERIODICITY.

→ LANTHANIDES AND
ACTINIDES.

One day → 4 lecture
cover

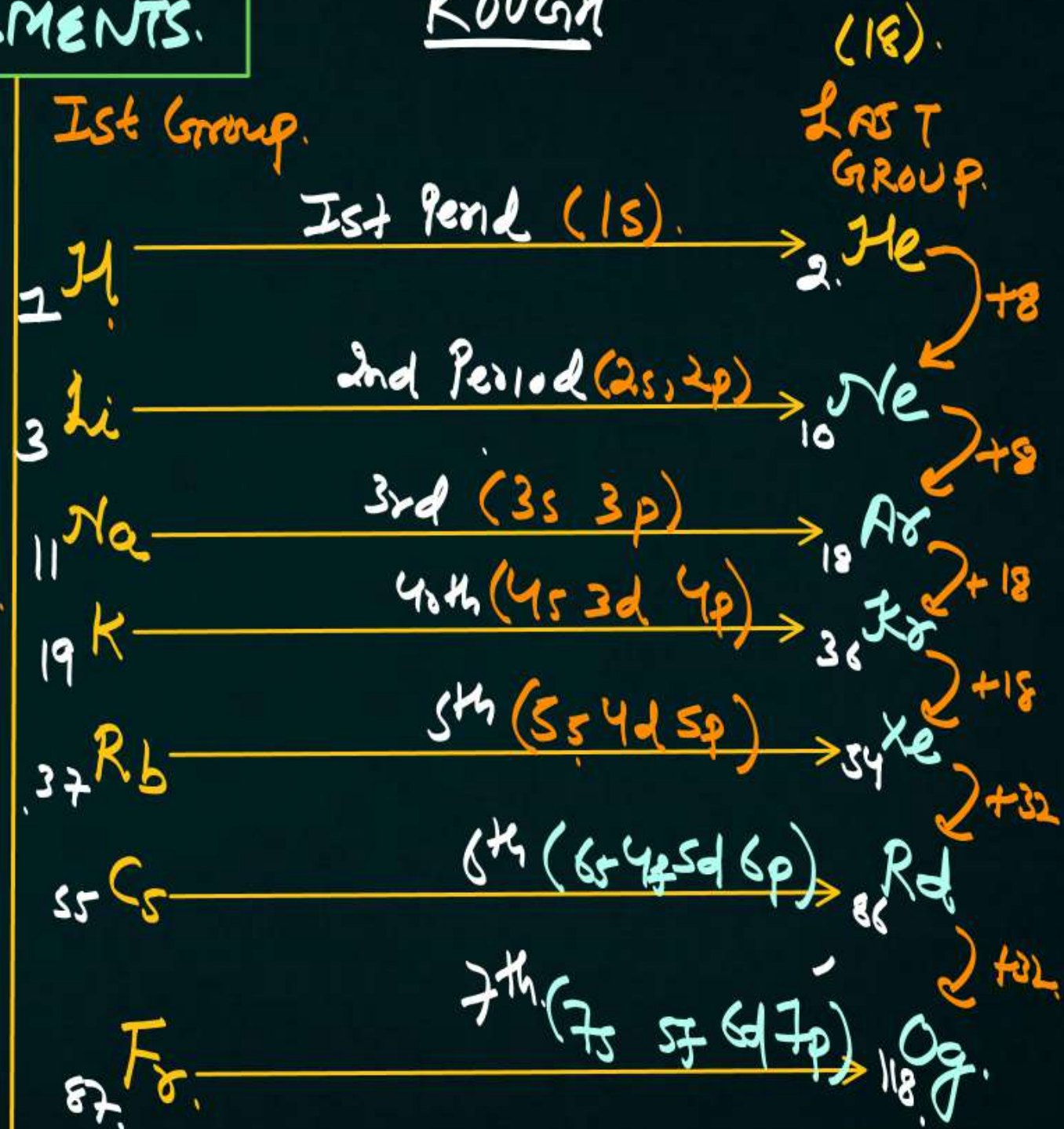
2 old + 2 live

→ Never miss the live class.

PERIOD WISE ELECTRONIC CONFIGURATION OF ELEMENTS.

ROUGH

- 1st Period has only two elements (H & He).
Starts from H & end at He, involves the filling 1s. (one orbital)
- 2nd Period has 8 elements. starts from Li (Z=3) to end at Ne (Z=10)., involves the filling 2s + 2p (4 orbitals)
- 3rd Period has also 8 elements, starts from Na (Z=11) to Ar (Z=18)., involving the filling to 3s + 3p (4-orbitals).
- 4th Period has 18 elements starts from K (Z=19) to Kr (Z=36).
- involves the filling of 4s 3d 4p. (9 orbitals).
- 5th Period also has 18 elements Rb (Z=37) to Xe (Z=54), involves the filling of 5s 4d 5p. (9-orbitals)



6th Period has 32 elements, starts from Cs (Z=55) + Rd(86).
involves the filling of 6s 4f 5d 6p (16 orbitals).

7th Period also has 32 elements, starts from Fr (87) to Og (118).
involves the filling of 7s 5f 6d 7p (16-orbitals).

#DIVISION OF ELEMENTS INTO BLOCKS.

1) S-BLOCK $\Rightarrow$ Elements in which last e^- enters in s-orbital of outermost shell or energy level. are called s-block elements

- General electronic configuration $\Rightarrow \underline{ns^{1-2}}$
- Involves two Groups (1 & 2).

1st-Group is also called Alkali metal (except hydrogen).
+ 2nd Group is also Alkaline earth metals.

1		2
ns^1		ns^2
H	1s ¹	
Li	2s ¹	Be 2s ²
Na	3s ¹	Mg 3s ²
K	4s ¹	Ca 4s ²
Rb	5s ¹	Sr 5s ²
Cs	6s ¹	Ba 6s ²
Fr	7s ¹	Ra 7s ²

2) P-Block : $\rightarrow$ Elements in which last e enters in p-orbital of outermost shell are p-block elements.

General electronic configuration : $\rightarrow ns^2 np^{1-6}$ (except He).

Involves the Group from 13 to 18. i.e.

Group no. 13. is also known as. Boron-family ($ns^2 np^1$)

Group no. 14. is also known as Carbon-family ($ns^2 np^2$).

Group no. 15, is also known as. Pnictogens ($ns^2 np^3$).

Group no. 16 is also known as CHALCOGENS ($ns^2 np^4$).

Group no. 17 is also known as. Halogen ($ns^2 np^5$).

Group no. 18 " " " " INERT GASES ($ns^2 np^6$)
or
NOBLE GASES except He ($1s^2$).

13	14	15	16	17	18
B	C	N	O	F	He
Al	Si	P	S	Cl	Ne
Ga	Ge	As	Se	Br	Ar
In	Sn	Sb	Te	I	Kr
Tl	Pb	Bi	Po	At	Xe
Nh	Fl	Mc	Lv	Ts	Rd
					Og

GROUP 1 2 ... 18

PERIOD 1 2 3 4 5 6 7

Alkali Metals, Alkaline Earth Metals, Transition Metals, Other Metals, Metalloids, Non-metals, Halogens, Noble Gases, Lanthanides, Actinides

Atomic Number, Symbol, Name, Average Atomic Mass

1	2											13	14	15	16	17	18
1	2											5	6	7	8	9	10
H	He											B	C	N	O	F	Ne
Li	Be											Al	Si	P	S	Cl	Ar
Na	Mg	3	4	5	6	7	8	9	10	11	12	Ga	Ge	As	Se	Br	Kr
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	In	Sn	Sb	Te	I	Xe
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	Tl	Pb	Bi	Po	At	Rn
Cs	Ba	57-71	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Th	Pb	Bi	Po	At	Rn
Fr	Ra	89-103	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

GROUP																		18							
1		2												13		14		15		16		17			
PERIOD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22			
1	1 H Hydrogen 1.008																	2 He Helium 4.003							
2	3 Li Lithium 6.94	4 Be Beryllium 9.012											5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01	8 O Oxygen 16.00	9 F Fluorine 19.00	10 Ne Neon 20.18							
3	11 Na Sodium 22.99	12 Mg Magnesium 24.31											13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.95							
4	19 K Potassium 39.10	20 Ca Calcium 40.08	21 Sc Scandium 44.96	22 Ti Titanium 47.88	23 V Vanadium 50.94	24 Cr Chromium 52.00	25 Mn Manganese 54.94	26 Fe Iron 55.85	27 Co Cobalt 58.93	28 Ni Nickel 58.69	29 Cu Copper 63.55	30 Zn Zinc 65.39	31 Ga Gallium 69.72	32 Ge Germanium 72.64	33 As Arsenic 74.92	34 Se Selenium 78.96	35 Br Bromine 79.90	36 Kr Krypton 83.79							
5	37 Rb Rubidium 85.47	38 Sr Strontium 87.62	39 Y Yttrium 88.91	40 Zr Zirconium 91.22	41 Nb Niobium 92.91	42 Mo Molybdenum 95.96	43 Tc Technetium (98)	44 Ru Ruthenium 101.1	45 Rh Rhodium 102.9	46 Pd Palladium 106.4	47 Ag Silver 107.9	48 Cd Cadmium 112.4	49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8	52 Te Tellurium 127.6	53 I Iodine 126.9	54 Xe Xenon 131.3							
6	55 Cs Cesium 132.9	56 Ba Barium 137.3	57-71 Lanthanides	72 Hf Hafnium 178.1	73 Ta Tantalum 180.9	74 W Tungsten 183.9	75 Re Rhenium 186.2	76 Os Osmium 190.2	77 Ir Iridium 192.2	78 Pt Platinum 195.1	79 Au Gold 197.0	80 Hg Mercury 200.5	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 209.0	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)							
7	87 Fr Francium (223)	88 Ra Radium (226)	89-103 Actinides	104 Rf Rutherfordium (261)	105 Db Dubnium (268)	106 Sg Seaborgium (271)	107 Bh Bohrium (278)	108 Hs Hassium (277)	109 Mt Meitnerium (276)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (280)	112 Cn Copernicium (285)	113 Nh Nihonium (284)	114 Fl Flerovium (289)	115 Mc Moscovium (288)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)							

78

Pt

Platinum

195.1

Atomic Number

Symbol

Name

Average Atomic Mass

78 — Atomic Number

Pt — Symbol

Platinum — Name

195.1 — Average Atomic Mass

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La Lanthanum 138.9	Ce Cerium 140.1	Pr Praseodymium 140.9	Nd Neodymium 144.2	Pm Promethium (145)	Sm Samarium 150.4	Eu Europium 152.0	Gd Gadolinium 157.2	Tb Terbium 158.9	Dy Dysprosium 162.5	Ho Holmium 164.9	Er Erbium 167.3	Tm Thulium 168.9	Yb Ytterbium 173.0	Lu Lutetium 175.0
89	90	91	92	93	94	95	96	97	98	99	100	101	102	103
Ac Actinium (227)	Th Thorium 232.0	Pa Protactinium 231.0	U Uranium 238.0	Np Neptunium (237)	Pu Plutonium (244)	Am Americium (243)	Cm Curium (247)	Bk Berkelium (247)	Cf Californium (251)	Es Einsteinium (252)	Fm Fermium (257)	Md Mendelevium (258)	No Nobelium (259)	Lr Lawrencium (262)

Q. Seventh Period involves the filling of

a) 7s 6d 7p

b) 7s 4p 5d 7p

c) 7s 5f 6d 7p

d) 7s 6f 5d 7p

s

s p

s p

s d p

s d f

s f d p

s f d f

$n s (n-2) f (n-1) d n p$

Q. GENERAL OUTER ELECTRONIC CONFIGURATION OF p-block elements.

a) $ns^{0-1} np^{1-6}$

b) $ns^{0-2} np^{0-6}$

c) $ns^2 np^{1-6}$

d) $ns^2 np^2$

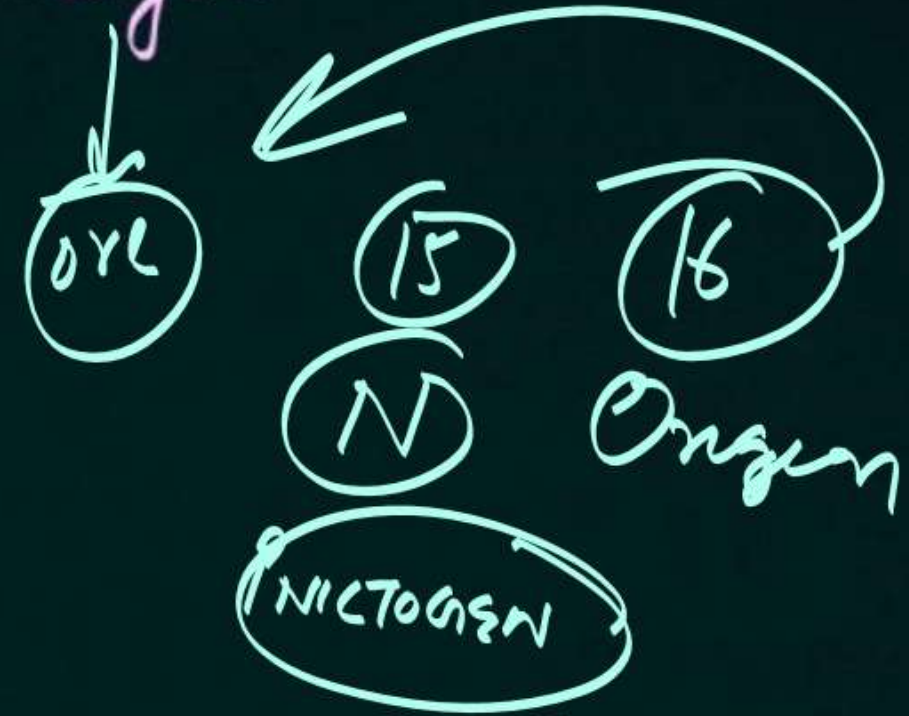
Q. Which of the following group elements are called chalcogens.

a) Group. 13

b) Group. 14

c) Group 15

d) Group 16.



Q. What is the atomic No. of Element $\rightarrow$ Og.

a) 110

b) 100

c) 101

d) 118.

Homework



Q. Write the all elements of s + p block with tricks?

COMMUNITY ME
POST KAR DENA
BHAICHARADN TOP.
VISHAL SIR ZINDABAD.

REVISION KAR K SONA.

Thank
You